BNCS

InChi

Digital Trade Platform Internship Program

University Collaboration Brief

Proposed Eight Week Programme

Singapore

Executive Summary

BNCS - Biz N Consumer Solutions Pte Ltd proposes an eight-week Digital Trade Platform Internship Program for selected Year 2 and Year 3 university students in Computer Science, Artificial Intelligence, Data Science, Cybersecurity, Software Engineering, and Product Design. Students will contribute to the controlled development of InChi, a digital trade gateway designed to support structured engagement between qualified Indian exporters and verified Chinese commercial demand.

The programme is designed as a supervised venture-build sprint rather than a conventional observation-based internship. Students will work in an integrated product team to create a bounded pilot platform, technical proof-of-concepts, tested workflows, quality evidence, and complete handover documentation.

The programme is structured to

- Translate a real cross-border trade workflow into a working digital product
- Develop a controlled internal pilot platform rather than unfinished production software
- Create reusable product, engineering, data, AI, security, and testing assets
- Give students practical experience in Agile product delivery and multidisciplinary collaboration
- Produce a demonstrable end-to-end supplier-to-opportunity workflow
- Prepare a documented foundation for future professional development

Core Principle
Students build a controlled pilot platform, reusable technical assets, and validated learning under supervision. They do not own production deployment, regulatory interpretation, or irreversible architecture decisions.

Programme Theme

InChi Pilot Gateway Build Studio

Duration	Eight weeks
Work commitment	Six working hours per day, five days per week
Mode	Remote or hybrid, as agreed with the university
Operating framework	Agile product sprint studio

Programme Domains

- Web product engineering
- Cross-border trade workflow digitisation
- Relational data modelling and API development
- AI-assisted supplier and buyer matching
- Cybersecurity and DevSecOps
- Quality assurance and test automation
- Product design and Agile delivery operations

Programme Objective

The cohort will develop InChi Pilot Gateway v0.1: a private, controlled web application that enables authorised BNCS users to register suppliers and products, assess China readiness, capture buyer requirements, review AI-assisted matches, track commercial opportunities, and view basic pilot performance.

The internship target is a supervised pilot MVP or, where technical evidence does not support MVP acceptance, a demonstrable functional prototype with complete technical handover documentation.

Pilot Platform Scope

The internship focuses on the operating spine required to demonstrate the InChi concept for one pilot sector. It does not attempt to reproduce a large public marketplace.

Core Modules

Module	Internship Build Scope
Supplier and Product Registry	Company profiles, product information, capabilities, certifications, capacity, export experience, and document-status metadata.
China Readiness Assessment	Structured questions, rules-based scoring, missing-information indicators, corrective actions, and human approval.
Buyer Requirement Capture	Product specifications, quantity, commercial parameters, certification needs, and verification status.
Matching Desk	Mandatory filters, weighted criteria, AI-assisted semantic comparison, ranked suppliers, explanations, and human decisions.
Opportunity Pipeline	Introductions, meetings, RFQs, samples, quotations, negotiations, purchase orders, and outcome tracking.
Management Dashboard	Supplier, buyer, matching, opportunity, activity, and conversion indicators.
Profile Preview	Controlled English and Mandarin-ready presentation of approved supplier and product information.

Team Structure

Recommended cohort: Eight interns

Role 1 Technical Project Coordinator and Product Operations Intern

Discipline: Computer Science, Information Systems, Engineering Management, or related discipline

Mission: Maintain delivery discipline and ensure requirements, decisions, blockers, and deliverables remain visible.

Responsibilities

- Maintain the sprint board, backlog, action log, risk register, and decision register
- Coordinate Monday sprint starts, Tuesday to Thursday stand-ups, and Friday sprint reviews
- Track dependencies, acceptance criteria, deadlines, and weekly demonstrations
- Prepare weekly status reports and consolidate the final handover

Deliverables

- Master project plan and sprint records
- Risk, issue, action, and decision logs
- Weekly status reports
- Final contribution and handover report

Role boundary: Coordinates the student team but does not approve architecture, security, or product scope.

Role 2 Product UX and Commercial Workflow Analyst Intern

Discipline: Computer Science, Product Design, Information Systems, Business Analytics, or related discipline

Mission: Translate trade processes into clear user journeys, interfaces, field definitions, and acceptance criteria.

Responsibilities

- Map supplier onboarding, buyer requirements, matching, and opportunity workflows
- Create wireframes and interface specifications
- Maintain user stories, field dictionary, and acceptance criteria
- Conduct usability testing and support user acceptance testing

Deliverables

- Workflow maps and user journeys
- Wireframes and interface specifications
- Field dictionary
- User acceptance test scripts

Engineering Roles

Role 3 Frontend Web Engineer Intern

Discipline: Computer Science, Software Engineering, or related discipline
Mission: Build a responsive and usable interface for the pilot platform.

Responsibilities

- Develop authenticated layouts, navigation, forms, lists, dashboards, and status components
- Build supplier, product, buyer requirement, matching, and opportunity screens
- Implement form validation and clear error states
- Create reusable UI components and responsive layouts

Deliverables

- Working application interface
- Reusable component library
- Frontend tests
- Frontend setup documentation

Role 4 Full Stack Integration Engineer Intern

Discipline: Computer Science, Software Engineering, or related discipline
Mission: Connect the interface, business logic, data services, authentication, and staging environment.

Responsibilities

- Implement application routes and authentication integration
- Connect frontend modules to approved services
- Implement role-sensitive workflows, error handling, and audit events
- Maintain the integrated staging build and resolve integration defects

Deliverables

- Integrated application
- Authentication and role workflows
- Staging build
- Integration and deployment documentation

Data and Intelligence Roles

Role 5 Backend and Data Engineer Intern

Discipline: Computer Science, Software Engineering, Data Engineering, or related discipline
Mission: Create the structured data foundation for suppliers, buyers, matching, and opportunities.

Responsibilities

- Design and implement the relational data model
- Create database migrations, validation rules, data services, and APIs
- Support search, filtering, interaction history, reporting, and synthetic seed data
- Maintain the data dictionary and backup guidance

Deliverables

- Database schema and migrations
- Data services and APIs
- Synthetic demonstration dataset
- Data dictionary and backup instructions

Role 6 AI Matching and Data Intelligence Intern

Discipline: Artificial Intelligence, Data Science, Computer Science, or related discipline
Mission: Build a human-controlled AI-assisted matching demonstration and evaluate its usefulness.

Responsibilities

- Create product and requirement taxonomies
- Implement mandatory filters and weighted matching rules
- Prototype semantic comparison and match explanations
- Evaluate match quality, false results, limitations, and non-AI fallback behaviour

Deliverables

- Rules-based matching baseline
- AI-assisted ranking prototype
- Evaluation dataset and report
- AI limitations and future-development brief

Role boundary: AI recommendations remain advisory and require authorised human approval.

Quality and Security Roles

Role 7 QA and Test Automation Engineer Intern

Discipline: Computer Science, Software Engineering, Artificial Intelligence, or related discipline
Mission: Provide objective evidence that the product works as specified.

Responsibilities

- Create test strategy, test cases, and requirements traceability
- Conduct functional, integration, regression, and end-to-end testing
- Develop selected automated tests and maintain the defect register
- Coordinate user acceptance testing and provide the final quality verdict

Deliverables

- Test plan and traceability matrix
- Automated regression suite
- Defect register
- Final QA and UAT report

Role 8 Cybersecurity and DevSecOps Engineer Intern

Discipline: Cybersecurity, Computer Science, Information Security, or related discipline
Mission: Establish basic security discipline and document what remains unsuitable for production.

Responsibilities

- Maintain the threat model, access matrix, and data classification rules
- Test authentication, authorisation, database policies, file handling, and secrets management
- Run dependency and configuration scans
- Support secure staging deployment, backup procedures, and security documentation

Deliverables

- Threat model and access-control matrix
- Security test and scan reports
- Deployment-security checklist
- Production-readiness gap report

Role boundary: Supports security testing but does not certify the platform as production secure.

Weekly Operating Rhythm

Each intern works six hours per day from Monday to Friday. Meetings are included within the six-hour working day.

Day	Ceremony	Duration	Purpose
Monday	**Sprint Start**	60 minutes	Confirm sprint goal, priorities, responsibilities, acceptance criteria, dependencies, and Friday demonstration.
Tuesday	**Stand-up**	15 minutes	Review completed work, planned work, blockers, decisions, and delivery confidence.
Wednesday	**Stand-up**	15 minutes	Review progress, integration dependencies, quality concerns, and required support.
Thursday	**Stand-up**	15 minutes	Confirm demonstration readiness and escalate unresolved blockers.
Friday	**Sprint Review**	90 minutes	Demonstrate working outputs, review evidence, record feedback, conduct retrospective, and prepare the next sprint.

Daily Work Allocation

- Primary assigned build or product work
- Integration, peer review, or paired work
- Testing and defect correction
- Documentation and project-tracker updates

Every task must have an owner, due date, acceptance criteria, evidence, and reviewer. Work is not complete merely because it has been submitted.

Eight Week Sprint Plan

Weeks One to Four establish the product foundation and complete the supply-side and demand-side workflows.

Week	Sprint Focus	Activities and Required Deliverable
1	**Product Definition**	Confirm user roles, workflows, MVP scope, data entities, architecture, security rules, repositories, sprint board, and test strategy. Deliverable: Approved product-definition pack and running application skeleton.
2	**First Vertical Slice**	Implement login, internal roles, supplier creation, product creation, saving, retrieval, editing, and initial security tests. Deliverable: Working authenticated supplier-to-product flow in staging.
3	**Supplier Registry**	Complete supplier onboarding, product catalogue, certification and document-status metadata, search, filtering, bilingual-ready fields, test data, and audit events. Deliverable: Demonstration supplier registry.
4	**Readiness and Demand**	Implement China-readiness questionnaire and scoring, corrective-action indicators, buyer profiles, buyer requirements, human approval, and access separation. Deliverable: Qualified supplier and structured buyer requirement ready for matching.

Week Four Review Gate

The team must demonstrate supplier onboarding, readiness assessment, buyer-requirement capture, data validation, and authorised access before the matching sprint begins.

Eight Week Sprint Plan

Weeks Five to Eight complete the intelligence layer, opportunity workflow, testing, demonstration, and handover.

Week	Sprint Focus	Activities and Required Deliverable
5	**AI Assisted Matching**	Implement mandatory filters, weighted criteria, semantic comparison, ranked suppliers, explanations, missing-information warnings, human decisions, evaluation cases, and rules-only fallback. Deliverable: Explainable matching demonstration.
6	**Opportunity and Dashboard**	Convert an approved match into an opportunity, record interactions, update commercial stages, calculate dashboard indicators, and display a controlled bilingual supplier profile. Deliverable: Complete supplier-to-opportunity walkthrough.
7	**Hardening and UAT**	Freeze features, conduct integration and regression testing, run user acceptance testing, correct defects, verify permissions, test backup and deployment, and prepare the release candidate. Deliverable: Tested release candidate and documented limitations.
8	**Demo and Handover**	Complete remediation, final demonstration, source-code packaging, setup instructions, product documentation, data dictionary, AI evaluation, QA report, security report, and continuation backlog. Deliverable: Accepted pilot MVP or classified functional prototype.

Feature Freeze

New feature development ends after Week 6. Weeks 7 and 8 are reserved for quality, security, usability, documentation, demonstration, and handover.

AI Matching Framework

The principal AI element is the InChi Match and Readiness Copilot. It assists authorised users in comparing structured supplier capabilities against buyer requirements. It does not approve suppliers or make commercial decisions.

Step	Stage	Function
1	**Mandatory Filters**	Product category, certification, production capacity, minimum order quantity, price range, export capability, and delivery requirements.
2	**Weighted Commercial Score**	Approved weights assess the degree of alignment across structured commercial criteria.
3	**Semantic Comparison**	AI assists with differently worded product descriptions, requirement interpretation, and multilingual terminology.
4	**Explanation and Gaps**	The system explains potential fit, identifies missing information, and displays limitations.
5	**Human Decision**	An authorised user approves, rejects, or defers the suggested match.

AI Evaluation

- Use synthetic or approved demonstration data
- Compare AI-assisted recommendations against reviewed test cases
- Record false matches, missed matches, confidence, and failure behaviour
- Maintain a rules-only fallback
- Require human approval before an opportunity is created

Innovation and Delivery Log

The Technical Project Coordinator maintains the Innovation and Delivery Log as the programme's institutional memory. The log converts weekly experimentation into reusable product and engineering learning.

Decisions Register

Decision | Date | Rationale | Alternatives | Approver | Implications

Experiments Log

Hypothesis | Method | Outcome | Evidence | Continue, revise, or stop

Failure and Learning Log

What failed | Cause | Incorrect assumption | Lesson | Future prevention

Risk and Issue Register

Risk or issue | Impact | Owner | Mitigation | Due date | Status

Carry Forward Register

Build next | Research next | Defer | Archive | Reusable assets

Week 8 final delivery includes the completed Innovation and Delivery Log, which supports product continuity, technical handover, and future development planning.

Programme Success Metrics

Success Measure	Required Evidence
Working vertical slice	Login, supplier, and product workflow completed by Week 2
Supplier demonstration data	At least 15 suppliers and 30 products
Buyer requirements	At least 10 structured demonstration requirements
Matching output	Ranked recommendations with explanations and human decisions
Commercial workflow	Approved match converted into a tracked opportunity
Quality evidence	Traceable test pack and final QA report
Security evidence	Threat model, access tests, scan results, and gap report
Handover	Reproducible setup, source code, product documentation, and continuation backlog

Evaluation Areas

- Quality and completeness of accepted work
- Reliability against commitments
- Technical and professional growth
- Problem-solving and responsible escalation
- Testing and documentation discipline
- Collaboration and response to feedback
- Final handover quality

Governance Rules

Students May Work On	Students May Not Independently Own
Pilot web application	Production release approval
Synthetic demonstration data	Live confidential commercial data
Prototype matching logic	Regulatory or legal decisions
Testing and security assessments	Autonomous supplier approval
Documentation and research	Irreversible architecture decisions
Staging environment under supervision	Unsupervised security certification

Working Rules

- Every task must have written acceptance criteria
- No feature enters the shared build without review
- Critical defects take priority over new features
- Blockers lasting more than one working day must be escalated
- No credentials may be stored in source code
- No confidential information may be used without written approval
- New feature development ends after Week 6
- Completion requires working evidence, testing, and documentation

Supervision Model

Programme Role	Responsibility
BNCS Programme Sponsor	Provides programme authority, commercial direction, weekly review participation, and final acceptance.
Product and Venture Lead	Defines product priorities, resolves scope questions, approves workflows, and accepts product outputs.
Senior Technical Mentor	Provides architecture guidance, code review, integration support, technical decisions, and engineering-quality control.
University Coordinator	Supports student participation, attendance, academic alignment, and university communication.
Technical Project Coordinator Intern	Maintains daily delivery coordination, sprint records, risk visibility, and programme documentation.

Team Composition Summary

Workstream	Count
Project and Product Operations	1
Product UX and Workflow	1
Frontend Engineering	1
Full Stack Integration	1
Backend and Data Engineering	1
AI Matching and Data Intelligence	1
QA and Test Automation	1
Cybersecurity and DevSecOps	1
Total	**8 interns**

Programme Positioning

This programme treats students as supervised product contributors. They will work across product design, engineering, data, artificial intelligence, quality assurance, cybersecurity, and Agile delivery within a real early-stage digital platform environment.

University Learning Outcomes

- Apply classroom knowledge to a multidisciplinary software product
- Work within an Agile sprint and demonstration cadence
- Translate business workflows into product and data requirements
- Develop, integrate, test, and document web-platform components
- Evaluate AI-assisted decision support responsibly
- Apply practical cybersecurity and data-handling controls
- Experience code review, user acceptance testing, and technical handover
- Understand the difference between a prototype, pilot MVP, and production system

Final Programme Outcome

At the conclusion of eight weeks, the cohort will present a working demonstration of InChi Pilot Gateway v0.1 or a formally classified functional prototype, together with complete product, technical, testing, security, AI, and handover documentation.

The final output will be evaluated on evidence, reliability, usability, learning, and reusability rather than the quantity of code produced.

Prepared by BNCS - Biz N Consumer Solutions Pte Ltd | Singapore